

Performance Evaluation of AODV and DSR Routing Protocols for Flying Ad-hoc Network using Highway Mobility Model

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Abstract: The Flying Ad-hoc Network (FANET) is a special type of mobile ad-hoc network that provides communications among Unmanned Ariel Vehicles (UAV). These UAVs can reduce human intervention to a great extent by giving numerous applications under different domains such as transportation, military, healthcare, traffic monitoring, surveillance, etc. In FANET, communication is relatively challenging due to their complex infrastructure, unspecified architecture, and rapid mobility of nodes. The work embodied in this paper is focused on traffic surveillance of highways using UAVs. UAVs can help in reducing the number of accidents by sharing real-time as well as the accurate status of highways among vehicles and the control station. On the other hand, they can also be used to track specific vehicles on the road. In FANET, both routing protocols and mobility models play a crucial role in this process of information exchange. In this paper, the comparison and performance evaluation of two well-known reactive routing protocols viz. Ad-hoc On-demand Distance Vector (AODV) and Dynamic Source Routing (DSR) have been carried out using highway mobility model for traffic surveillance in FANET environment. Implementation of both protocols has been tested on several traffic patterns, mobility, and varying network loads. Both AODV and DSR enable significant performance variations; however, they share on-demand behavior. Packet delivery fraction, average end-to-end delay, normalized routing load, packet loss, routing overhead, and throughput are used to analyse both protocols performance. Based on experimental analysis using NS-2 under constant bit rate (CBR) and TCP traffic sources, it can be stated that AODV outperforms DSR in almost every aspect.

Keywords: *Flying Ad-hoc Network (FANET), Unmanned Ariel Vehicles (UAV), Ad-hoc On-demand Distance Vector (AODV), Dynamic Source Routing (DSR), Constant Bit Rate (CBR), Transmission Control Protocol (TCP), Highway Mobility Model*

1. Introduction

In FANET, flying drones or UAVs act as hubs that can move anywhere in the air and make communications with each other [1]. There is no need for a separate physical communication medium to communicate, as shown in figure 1. Few nodes in FANET can act as cluster heads that may communicate with satellite networks and other nodes in the network. UAVs are enabled with the

communication system, the video camera, and sensor. Nodes in FANET are connected in ad-hoc mode and are generally considered as the special form of Vehicular Ad-hoc Network (VANET) and Mobile Ad-hoc Network (MANET) [2]. FANET nodes are small UAVs that are very reasonable in cost, quick to organize, and adaptable in diverse network circumstances. These points make FANET and its nodes (UAVs) extremely interesting for many applications, viz. traffic monitoring, environment sensing, healthcare, military, etc. [3]. Routing in the FANET environment is challenging and needs to be dealt with utmost care and optimization. In optimizing the performance of routing protocols of FANET, the mobility model is also playing a significant role. Thus, in this paper, two famous reactive UAV routing protocols viz. AODV and DSR have been examined using the highway mobility model. The motivation behind finding out the best routing protocol for the highway mobility model is to overcome the limitations of traditional traffic surveillance and monitoring. Nowadays, FANET enables traffic monitoring and surveillance efficiently and provides a wider variety of applications at a very low cost. With the help of FANET, one can easily monitor the traffic situations of a city/ highway at a single place of control that was not possible earlier in conventional traffic monitoring. UAVs in FANET can efficiently accomplish such tasks with reduced time as well as other resources. In general, a considerable workforce and resources are required to monitor the actions of a town. Figure 2 represents the general scenario to monitor the traffic activities in a city using UAVs.

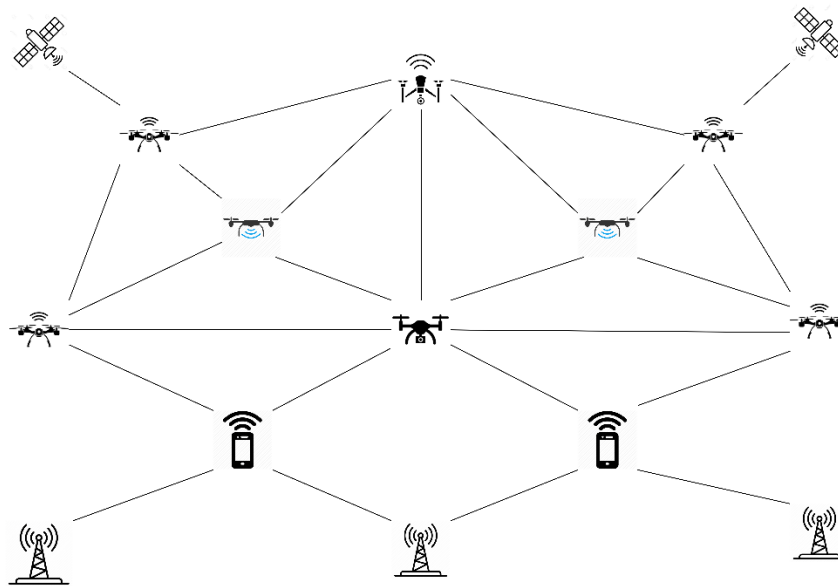


Fig. 1: Illustration of a Flying Ad Hoc Network along with base station and satellite communication

UAVs are capable of monitoring traffic on highways and can monitor several activities in narrow and congested streets, where human intervention is relatively complex on an urgent basis. The main advantage of such traffic monitoring systems is that they do not demand complex infrastructure and several pre-requisites (i.e., training) because they are easy to operate, lightweight, and convenient to use. Multiple UAVs of a city can be connected via a wireless network for making a FANET, thus producing real-time monitoring at a single place[4]. FANET architecture of such a network can vary from simpler to complex structure depending upon the layout of roads and the city [5]–[11].

Besides several advantages, these UAVs suffer from some drawbacks also, i.e., creating air traffic, exposure to physical damages, getting injured with hobby drones, irritate citizens, etc. Even traffic

police can take advantage of the FANETs to provide real-time traffic updates to the citizens to reduce traffic jams. To enforce law and order during critical situations, FANETs can record photographs and videos, which may be further used as proof to track and control offenders [12]. This wider range of applications motivated us to work towards finding out efficient routing algorithms for the FANET environment so that the overall task can be faster and optimized. In general, the major contributions of the paper are highlighted as follows:

- To simulate and compare AODV and DSR routing protocols on the FANET environment using various performance metrics.
- To monitor the traffic and the workforce using the highway mobility model.



Fig. 2: Traffic monitoring and control in a city through UAVs[13]

The rest of this paper is organized as follows. Some of the related work with implementation and analysis of FANET routing protocols using different mobility models has been briefed out in section 2. In contrast, the working of AODV and DSR along with the classification of UAV routing protocols are presented in section 3. Section 4 provides a brief introduction about mobility models i.e., random waypoint and highway mobility model. The overall simulation process along with metrics used for performance analysis is discussed in Section 5. Experimental results of the simulation are analyzed and presented in section 6. Finally, section 7 concludes the study with some future directions.

2. Related Work

Several implementations and performance analysis of FANET routing protocols have been carried out in research literature using various traffic sources on different mobility models along with other factors such as the number of nodes, speed of nodes, network load, etc. This section will brief some of the work related to the implementation and analysis of FANET protocols using different mobility models and network configurations.

Singh and Verma *et al.* [14] have presented an implementation of Optimized Link State Routing Protocol (OLSR) in FANET and investigated the performance analysis under several mobility models viz. reference point group mobility (RPGM), random waypoint, Manhattan-Grid, and Pursue. For conducting an experimental study, they had fixed the simulation area to $400 \times 600 m$ using NS-2. Based on the experimental analysis, optimum performance of OLSR was observed in Pursue mobility model. FANET routing protocols can be classified into different classes. In this regard, Tareque *et al.* [15] classified these routing protocols into six major categories viz. static, reactive, proactive, position/

geographic-based, hierarchical and hybrid. Using several performance criteria, they have examined and compared these routing protocols so that one can decide the suitability according to their requirements and scenarios for establishing FANETs. On the other hand, Arafat and Moh [16] presented a detailed survey of UAV routing protocols. They have presented several UAV architectures, mobility models, and routing protocols in a structured manner.

Vasiliev *et al.*[17] presented the performance comparison of OLSR, AODV, and Hybrid Wireless Mesh Protocol (HWMP) for analyzing the Quality of Service (QoS) in FANETs. For simulating the group behavior of UAVs, their simulation setup consists of a varying number of nodes (10, 15, and 20) for each type of protocol (i.e., proactive, reactive, and hybrid) with Gauss-Markov as mobility model. Analysis of these protocols has been carried out using Packet Delivery Ratio (PDR), hop-count, and network overhead performance metrics under the variable size simulation area (from 250 to 750m). Based on the experimental analysis, they have observed higher PDR using HWMP as compared to the other two protocols under consideration. On the other hand, Singh and Verma [18] analyzed the performance of three routing protocols viz. Destination Sequenced Distance Vector (DSDV), OLSR, and AODV routing protocols using random waypoint mobility model. Average throughput, end-to-end delay, PDR, and speed of mobile nodes have been considered for analyzing the performance of these routing protocols. Based on the experimental analysis, they have observed higher performance in OLSR than DSDV and AODV in the FANET environment.

Maakar *et al.*[19] have analyzed the performance of OLSR and AODV routing protocols for a 3D FANET environment based on the Gauss-Markov Mobility Model. For experimental analysis, their simulation environment consists of a varying group of UAVs with several packets, and a cubical area of $1700 \times 1700 \times 1500$ m has been considered with the Gauss-Markov model using NS-3 simulator. Average end-to-end delay, PDR, and throughput have been considered as performance metrics for the study. Simulation results and theoretical analysis stated that OLSR outperforms AODV in terms of PDR and real-time delivery of data.

Anand Nayyar [20] has presented the performance analysis of several routing protocols viz. AODV, DSDV, DSR, OLSR, AOMDV, HWMP for FANET environment by generating traffic source using CBR. The random waypoint mobility model was considered as the mobility model for all experimental analyses using NS-2 simulator. Based on the simulation, it was observed that the performance of OLSR and HWMP was found suitable in dynamic environments of FANETs. On the other hand, performance analysis of the FANET 3D environment based on the number of UAVs and a varying number of packets has been carried out in [21] using the AODV routing protocol and Gauss-Markov mobility model. The experimental analysis has shown a variation in the performance of AODV due to variation in both UAVs and packets. The major challenge in analyzing the performance of FANET is the change in network topology as compared to MANET and VANET [22].

Han Peng *et al.* [23] have proposed an algorithm that observes the unseen mobility profiles of the object and hence generates new classes. UAVs play an important role in applications where human intervention is not significant such as disaster management and search and rescue procedures. These UAVs use autonomous networks for their work; hence they do not require ground connection and communication. Once the UAVs acquire information about their neighboring nodes such as mobility, sensing, etc., it supports their decision-making process. Hence to smoothen the working of control and communication protocols, an unsupervised online learning algorithm has been discussed for UAVs to predict joint mobility along with profiling the object. Besides classifying the flying objects into various classes based on mobility, the solution also indicates their next locations. This method has broad applicability in unknown FANETs with heterogeneous nodes since it works with anonymous profiles of mobility. Motion trajectories of the object have been used to predict the joint mobility and profiling

of the object. The future locations, along with the forces driving the object are predicted with the help of the state transition equations when they are processed in terms of trajectories of motion. A hierarchical generative model is created for the forces (both direct and rotational) which are exerted and helps to classify the objects based on their mobile properties. Since the proposed method is unsupervised; hence it does not require any previous data for predictions, and neither is fixed for any specific class of objects, which makes it suitable for practical applications. For a reasonable amount of noise level, a success rate of 90% is proposed for profiling the objects, which is 7% more than the current state-of-the-art method. Also, by processing the trajectories of the object accumulated over time, the mobile properties of the object are tuned by the proposed novel online self-tuning algorithm.

Bowen Zeng *et al.* [24] have proposed a novel strategy for effective communication using dual antennas, which uses the predicted location along with the traced target to catch and trace the targets of communications. MANETs are one of the existing protocols of communication; still, they cannot be used for FANETs due to their configuration, which has an impact on the distance of communication and the high degree of mobility amongst the nodes. So, to resolve this issue, a hybrid omnidirectional and a multi-beam directional antenna structure is used for controlling the separation of data using a gradient elastic approach. This structure also resolves the directional deaf problem. The proposed solution shows that the robustness of the data link has increased by 35.8% in scenarios of high mobility. Also, the proposed solution of communication in dual antennas can transmit the data in FANETs at a high rate and stable fashion.

Nitesh Saxena *et al.* [25] have proposed decentralized node admission which is one of the most important and necessary security services in ad-hoc networks. It helps to deal with security services and primitives like secure coding and key management respectively along with dynamic membership, topology, and bootstrap. Since the MANET nodes are limited in resources, there must be minimal interaction between them to avoid unstable connections. This leads to efficient communication as well as computation. But the earlier proposed admission protocols require a large amount of interaction between the MANET nodes, besides being expensive. Here, an entirely non-interactive admission technique is offered for a temporary MANET that is efficient and secure. This protocol is based on bivariate polynomials that help in secret sharing among the nodes by treating them as private keys. Also, an efficient method is proposed, which allows for secure communications in flying devices.

Nishat I Mowla *et al.* [6] have proposed a reinforcement-based jamming attack strategy, which is adaptive and is used to develop a model-free Q-learning mechanism with an adaptive exploration-exploitation epsilon-greedy policy. UAVs use FANETs for communication which is decentralized in nature, but due to their wireless and unique nature of properties of the network, it is always at risk of a jamming attack besides other challenges. There are some of the challenges faced in the existing solutions as follows:

- Due to communication and power constraints in FANET's a centralized knowledge base cannot be used to make efficient decisions.
- The distributed mechanism of jamming becomes inadequate due to the presence of new spatial environments which contain unbalanced data as the UAVs in FANET have low density and high mobility.
- Imprecise probability estimation of the transitional probabilities is challenging in newly explored spatial environments while taking actions in the model-based jamming.

The new results of the above mechanism when simulated revealed that it outperformed the base methodology by reducing the number of hop-counts in the route locations of the jammer by a significant amount. The federated jamming detection mechanism achieved 39.9% higher average

accuracy than the one achieved by the NS-3 simulated FANET and standard CRAWDAD jamming attack dataset.

Stefano Rosati *et al.* [26] have compared the performance between OLSR and P-OLSR. FANETs are affordable, flexible, and easily deployed, thus having applicability in civilian and military applications. These networks comprise UAVs, and in this paper, the self-organization of these networks has been experimented. Even though these networks have dynamic topology as compared to MANETs and other standard vehicle routing protocols, establishing a communication network between the UAVs is a complicated process due to high mobility between the nodes. This leads to a failed result of tracking network topology for MANETs when the existing protocols of routing are used. Here, two different routing algorithms are used for the Ad-hoc networks: optimized link-state routing (OLSR) and predictive OLSR (P-OLSR). P-OLSR is designed for FANETs specific routing and is an extended version of OLSR. It is implemented in Linux and uses the Global Positioning System (GPS) for its working. Both media access control (MAC) layer emulations and real-world experiments are used to obtain and compare the results. Two autonomous fixed-wing UAVs, along with a ground node are used for testing the experiment and routing performance along with the communication range. On evaluation, it is concluded that the P-OLSR surpasses the performance of the OLSR in both the emulation and experimental results when the topology of the network changes frequently.

D. S. Lakew *et al.* [27] have discussed the existing protocols of routing for FANETs along with the application architectures, communication, and classification of the UAVs. Besides this, the performance of the existing routing protocols of FANETs is calculated along with the emphasis on the strengths, weaknesses, and features of different mobility models. Also, the existing challenges and open research issues are highlighted. As compared to a single extensive UAV system, the multiple UAV arrangement in an ad hoc fashion called FANET is used to finish complicated tasks collaboratively and cooperatively. But FANETs also introduce certain disputes in communication designs like routing due to the unique characteristics of the network i.e., high mobility, high frequency of the topology, and low density of the nodes. Hence the requirements of routing in the case of these networks outperform both VANETs and MANETs.

Ashish Khanna *et al.* [28] considered the problem of mutual exclusion in FANET. The proposed algorithm ensures mutual exclusion in FANETs, which is a token-based algorithm named as Mobile Resource Mutual Exclusion (MRME) algorithm. As FANETs topology is highly dynamic and fault-prone, the proposed technique handles this issue efficiently. This algorithm is fault-tolerant and scalable. Token-loss has been dealt with effectively to make the algorithm fault-tolerance. Here, link updating, token loss, throughput, and response time are independent of the number of users.

Jianmin Liu *et al.* [29] proposed an algorithm that represents a novel Q-learning-based protocol that is based upon a multi-objective optimization routing protocol for FANETs. To select a more reliable hop, the proposed method re-estimates the routing decision. The learning rate is adjusted by one-hop delay, and the mobility of neighbors changes the discount factor in the adjacent time intervals. The authors also proposed a new technique of exploitation and exploration in Q-learning routing. Chakraborty *et al.* [30] also presented an optimal traffic management scheme for heterogeneous networks. For improving quality of service in the heterogeneous networks a novel routing is presented in this work followed by experimental and theoretical analysis.

Kirshna Kumar *et al.* [31] have presented a novel drone-assisted distributing routing framework in IoT environments (D-IoT). QoS and drone mobility models are crucial factors for designing a drone-assisted distributed routing framework. For efficient route selection, a Neuro-fuzzy interference system is deployed. This system is consisting of five different layers i.e., fuzzy layer, T-norm layer, normalized layer, de-fuzzy layer, and an aggregated Output layer. To optimize network centric QoS

parameters focusing on route delay, residual route load, expected link availability period, and relative velocity of drones have been used. This drone-assisted ad-hoc networking can be used in various fields such as traffic monitoring, border monitoring, and the agriculture sector as well.

Cognitive radio network technology has emerged as a promising solution for many critical problems such as in UAVs where only one radio is used for multiple purposes such as energy, cost optimization, and size-constraint. Peng Feng *et al.* [32] gives a glimpse of multi-channel cognitive MAC protocol (CogMOR-MAC), based upon the Multi-channel Opportunistic Reservation (MOR) mechanism. This MOR protocol can reduce frequency hopping and primary user activities. This CogMOR-MAC protocol is beneficial in multi-UAVs, ad-hoc networks, and mobile ad hoc networks. CogMOR-MAC can provide reliable communication services between service units with only one radio. This mechanism is very effective in reducing collisions and waiting time.

Cong Pu *et al.* [33] have presented a jamming-resilient multipath routing protocol called JarmRout. The protocol is mainly based upon three factors i.e., traffic load, link quality, and spatial distance. The protocol can avoid intentional jamming, network failure (isolated and localized failures). In the proposed protocol, the packet delivery ratio, and packet delivery latency has been improved to some extent. The protocol is useful in enhancing the network resiliency despite the occupancy of malicious jammers in the network.

Aadil, F. *et al.* [34] have proposed an Energy-Aware Link-based Clustering (EALC) algorithm to deal with two main issues in FANET viz. limited energy, and inefficient routing. The proposed algorithm is an advanced variant of the K-mean clustering algorithm. In K-mean clustering, there is only one parameter used, i.e., degree of the neighborhood. In contrast, in EALC, two additional parameters are used i.e., energy level, and for selection of optimal cluster head (CH), distance to the neighbor is used. EALC is effective as it enhances the cluster life span, and energy utilization by saving the node energy. As compared to other AI algorithms such as Ant-Colony Optimization and Grey-Wolf Optimization, it performs better in energy consumption, cluster binding time, and cluster lifetime. On the other hand, Sinwar *et al.* [35] also presented the usage of ant colony optimization on DSDV, AODV, and AOMDV routing protocols for shortest paths in mobile ad-hoc network environments.

Sang, Q. *et al.* [36] proposed the state-of-art survey of various routing protocols presented in recent years in FANET. The authors have presented a comprehensive analysis of each protocol in terms of its characteristics and service quality. Oubbati OS *et al.* [37] proposed the ECaD algorithm, which predicts the link failure before their occurrences and balanced energy consumption among UAVs. In case of a path failure, the proposed approach maintains an additional alternative route. On the other hand, Khan MA *et al.* [38] presented a survey based on topology, working, and the limitations of the FANET environment. The authors also considered the factors affecting the performance of FANET i.e., traffic density, packet forwarding mechanism, routing overhead, and network mobility. For performance measurement throughput, delay, and load metrics are used. Oubbati OS *et al.* [39] proposed a novel survey that provides a complete analysis of important FANET routing protocols. The authors also discussed different constraints and limitations based upon important parameters. Various routing techniques and simulation tools are also described in this paper. Darabkh KA *et al.* [40] discussed recent advances in gadgets, mobile computing, and wireless communication that have made it possible to make convenient gadgets with high conveying and handling abilities than at any other time. In this article, the author proposes a Multi Data Rate Mobility Aware (MDRMA) convention in which novel routing and power-controlled systems have been recommended. MDRMA-Power controlled calculation technique includes a few changes in 802.11b norm, explicitly on collision avoidance method. The MDRMA-routing algorithm works on request path route establishment. One of

the significant discoveries of this protocol is to empower the routing layer and MAC sub-layer to exchange data to use the most elevated realistic information transmission rates for accomplishing quick and productive information traffic forwarding.

Due to the rapid advancements in 5G technology, FANET's are also capable of efficiently sharing of data using Blockchain-based approaches [41], [42]. On the other hand, for satellite-based communications, the collisions must be detected at an early stage [43], [44] for substantial random accesses. Security is one of the main concerns for both ad-hoc and IoT communications [45]–[47]. For efficient and secure mobile communication in FANET environment, advance cryptographic approaches must be developed. Several other implementations based on ad-hoc, vehicular and IoT based secure communications are presented in [48]–[51].

Consequently, the implementation of optimized routing in FANET is relatively complex [52], [53]. Several routing protocols have been developed so far, and the search for an optimum one is still going on. Though FANET provides versatile applications in real-life until now no major significant contribution in the field of FANET routing protocols has been reported because of the high mobility of nodes. In this regard, the performance analysis of existing AODV and DSR routing protocols for the FANET environment under several traffic patterns for highway mobility has been carried out. This work aims to find out optimum routing protocol by considering the high mobility behaviour of UAVs on highways so that the same can be utilized for traffic monitoring and surveillance.

3. Routing Protocols

Several routing protocols exist in the research literature for Ad-hoc networks. This section will be focusing on two famous reactive protocols for the FANET environment because of their high performances. Arafat and Moh [16] presented the detailed architecture for UAV communication networks along with the classification of UAV routing protocols. Part of their classification is presented in figure 3 for understanding the basic topology-based classification of UAV routing protocols. In topology-based routing, the protocols are mainly classified into four main categories viz. static, proactive, reactive, and hybrid.

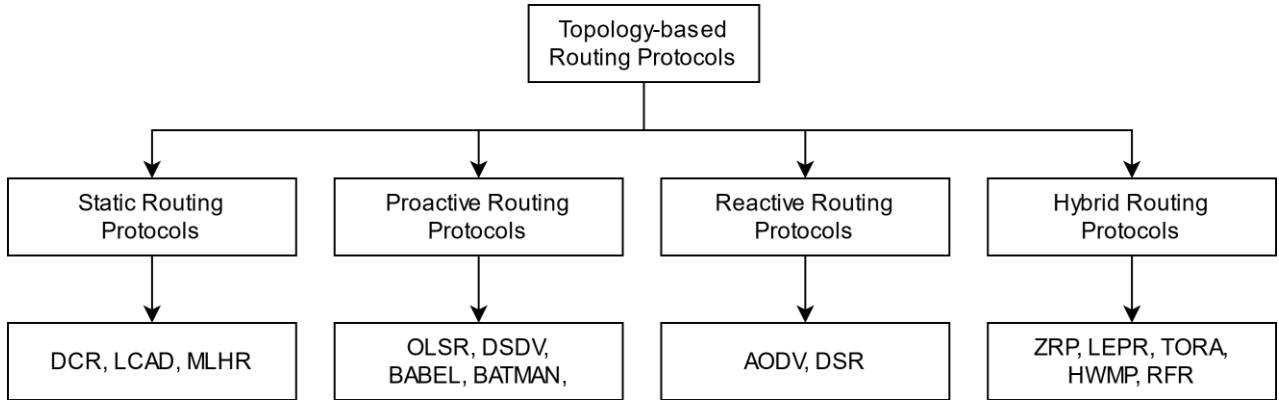


Fig 3. Routing protocols for Unmanned Aerial Vehicles [16]

3.1 Ad-hoc On-demand Distance Vector (AODV)

As the name suggests, the (AODV) [54], [55] routing protocol does not maintain routing information in advance; instead, it establishes routing information based on demand/ requirement. Due to this

nature, this protocol is named as on-demand routing protocol. The principle mechanism of this protocol is based on the distance vector routing approach, where each node maintains a routing table based on routing information shared by its neighbors called next hops. The next-hop further holds routing information about its neighbor on the target path. The minimum distance from one node to another node is calculated and updated on corresponding routing tables on a requirement basis. A routing record will be deleted if it has not been reactivated or used for a predetermined period. Moreover, it uses a sequence number-based mechanism to ensure the availability of a loop-free network all the time. For discovering and maintaining routes, AODV incorporates four control messages as follows:

1. Route Request (RREQ): The RREQ message is used to request a route when a node has no route to the destination node. The message is generally broadcasted to the network. Based on the reachability of RREQ to either target or mediator node, a route is determined. RREQ message includes several parameters such as broadcast ID, source address, target address, source node sequence number, and the last seen sequence number of the target node as given in (1):

(source_addr, source_seq_num, broadcast_id, dest_addr, dest_seq_num, hop_count) (1)

For the unique identification of RREQ, the pair of *source_address* and *broadcast_id* is used. On every new request, the *broadcast_id* is incremented one by the source. All neighbors in the network may send route reply (RREP) messages back to the source or broadcast RREQ to neighbors.

2. Route Reply (RREP): In response to RREQ, the RREP message is unicasted back to the originator node by any node that has a new route to target or target itself. Symmetric connections are used for RREP messages and reply messages pursue the inverting path of the respective route request message. After getting the RREQ message, each moderate node along the data path refreshes its table records for the target node. The intermediate or the destination node sends back the RREP message in case no route is present from that node to the source node and RREQ is unprocessed. The contents of the RREP message are mentioned in (2) as follows:

(source_addr, dest_addr, dest_seq_num, hop_count, lifetime) (2)

During transmission of RREP back to the source, a forward pointer has been setup by each node along the path. If a node is not on the path from the destination to the source where the RREP is traveling, the RREP times out after 3000 msec.

3. Route Error (REER): The route error message is used to broadcast the information about broken links when one or more destination nodes become unreachable from the sender node. At a point when a node finds a connection separation, it broadcasts a REER message to its neighbors, which in turn spreads the error message to the network especially towards those nodes whose data paths might be influenced by the disengaged connection.

4. HELLO: Hello as a greeting message is being broadcasted to check the network connectivity with neighbor nodes. This message is also known as a link status message because it provides various information about the node as well as the path/link in return.

3.2 Dynamic Source Routing (DSR)

DSR is another famous reactive routing protocol in which nodes find an optimal data path according to data transfer demand [56], [57]. For legitimate usage of the transmission capacity, the methodology to discover a route is possibly executed whenever a node demands a route. In this protocol, the source node performs a source routing to find out the entire path from the source node to the target node and inserts the addresses of all intermediate nodes into the packets. Different packets may follow different paths from the same source to destination; that is why it has been referred to as dynamic source routing.

The basic idea of DSR was introduced for ad-hoc networks where the width of the network is within the range of 5 and 10 hops, and movement of nodes is relatively at moderate speed. Concerning FANET, DSR uses both route discovery and route maintenance work together and are executed only when there is a demand:

1. Route discovery is needed for finding out the optimal route between a source node and a destination node. A route cache is being maintained by each node participating in the network to hold the information about the learned routes. When the source wants to send a packet to the destination, then the route is first checked from the route cache, if the route is found from the cache, then the source uses this route for sending packets to the destination otherwise the route discovery is utilized.
2. In contrast, route maintenance is used to maintain a loop-free path that ensures optimality even when there is any change in route. Also, it ensures the continuous monitoring and availability of routes from source to destination. If any hop is moved out or not working along the route, then the route cannot be utilized for transmission of packets and in that case, the route discovery needs to be executed again to maintain the route.

Like any other protocol, the DSR does not broadcast frequent messages. Due to this nature, DSR generates a minimal number of overhead packets.

4. Mobility Model

One of the main properties of nodes in the FANET environment is their mobility. In FANET, nodes are free to move in the surroundings as per requirements [58]. Among several mobility models, the highway mobility model provides the facility to mobilize the nodes in a highway environment. This type of mobility model has several real-life applications such as monitoring highway traffic through UAVs that are flying over highways, tracking the status of vehicles on roads, etc. In this model, operations of UAVs are designed in such a way that they should follow a predefined dedicated lane as of highway. The speed of nodes is based on their capabilities as decided during manufacturing. For experimental evaluations, varying speed of nodes has been considered in this work. The topography of a sample highway mobility model generated from Network Animator, demonstrating the movement of 50 UAVs is depicted in figure 4. The main characteristic of this model is the arrangement of nodes as

freeways on the map. Each freeway is designed in the form of a highway with bidirectional traffic flow i.e., upward and downward. The highway mobility model is based on the Random Waypoint mobility model [59], so both models are hereby discussed for ease of understanding.

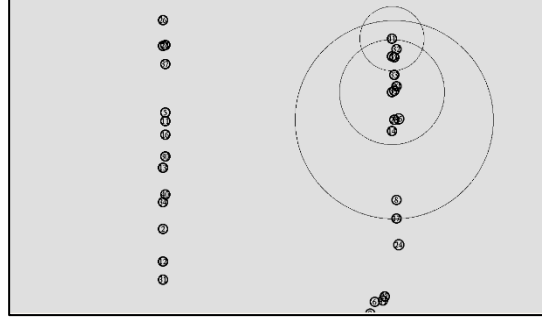


Fig.4: Topographical representation of 50 UAVs flying on a road for simulating highway mobility model

4.1 Random Waypoint Model:

In Random Waypoint, each node chooses the destination point randomly and moves towards it with a random velocity from the range of $[0 - v_{\max}]$. When our goal is achieved, then for a while, the node stops i.e., called pause time. After an interval, the node selects a destination point again in a random fashion. The entire process repeats until the whole simulation time gets over. The first step towards implementing a random waypoint model starts by selecting the parameters. Here each node is represented as a vector $P_i^{(j)}$, in which i denotes the movement duration of node and j which identifies the node itself. The movement trace of a node in the Random waypoint model [59] is as given in (3) as follows:

$$\{P_i^{(j)}\}_{i \in N_0} = P_0^{(j)}, P_1^{(j)}, P_2^{(j)}, P_3^{(j)}, \dots \quad (3)$$

If the node chooses random speed V_i from P_{i-1} to P_i and chooses pause time $T_{p,i}$ at P_i , the complete movement of the node can be seen as given in (4):

$$\{(P_i, V_i, T_{p,i})\}_{i \in N} = (P_1, V_1, T_{p,1}), (P_2, V_2, T_{p,2}), (P_3, V_3, T_{p,3}) \dots \quad (4)$$

4.2 Highway Mobility Model:

As compared to the Random Waypoint mobility model the nodes in the highway mobility model are restricted to operate on their lanes only. In highway mobility, the maximum velocity of a node is temporally dependent on its existing velocity as given in (5),
Formally,

$$|\vec{v}_i(t+1)| = |\vec{v}_i(t)| + \text{random}() \times |\vec{a}_i(t)| \quad (5)$$

On the other hand, if two mobile nodes are moving on the same highway lane and are within the specified Safety Distance (SD) measure, the velocity of the following node cannot go beyond the velocity of the preceding node as given in (6). Formally,

$$\forall i, \forall j, \forall t, D_{ij}(t) \leq SD \Rightarrow |\vec{v}_i(t)| \leq |\vec{v}_j(t)| \quad (6)$$

where if j is ahead of i in its lane.

Due to these characteristics, relationships among nodes in highway mobility are anticipated to follow spatial dependence. The same can also impose a node to follow strict geographic movements without changing the designated lane.

5. Simulation Process

UAVs in FANET use 802.11n MAC protocol and communicates on a 5GHz frequency band. Most of the work in FANET is based on the Random waypoint model, but for the societal point of view like highway traffic one must use a dedicated model that can simulate the behavior of highway mobility. This section discusses the simulation of the highway mobility model that is based on the Random waypoint model. As discussed in section 1, the scenarios presented in this paper uses two famous routing protocols i.e., AODV and DSR, and a new mobility model, i.e., highway mobility model for the same. For the practical simulation, two highways and three lanes are used to move nodes one after another. Highway mobility model is generated by using a Java-based software named IMPORTANT [60]. Overall simulation is carried out using well-known Network Simulator-2 (NS-2), IMPORTANT framework, traffic generators, and other tools as depicted in figure 5.

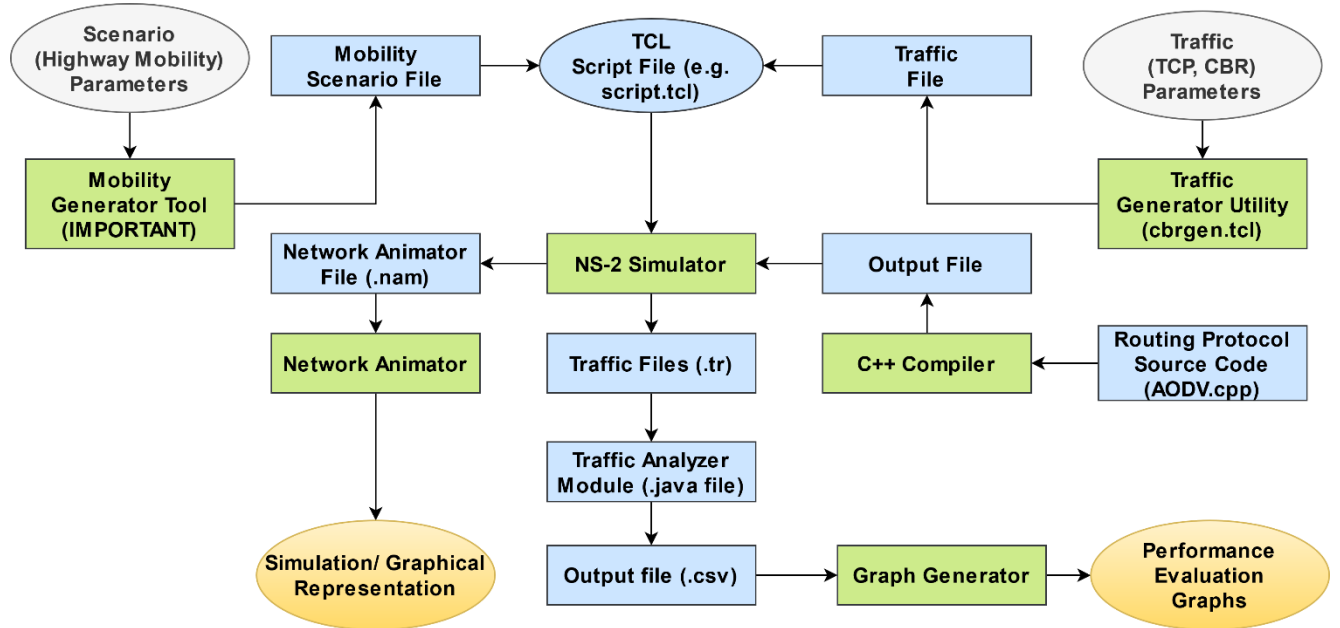


Fig.5: Flow Diagram of the overall simulation process

For simulating the behavior of flying ad-hoc networks and the highway mobility model, and carrying out experimental evaluations, Network Simulator-2 (NS-2) with Red Hat environment is being used. The NS-2 is a very powerful and famous discrete event-driven simulator [61], [62] developed at UC Berkeley. NS-2 provides various features such as simulating wired/ wireless networks, designing new protocols, simulating several networking protocols, evaluating traffic performances, etc. For simulating the highway mobility model, the ‘Mobility Generator’ tool is being used that is built upon ‘IMPORTANT’ (Impact of Mobility on the Performance Of Routing protocols in Ad-hoc NeTwork) framework [60]. In this work, the authors have used two highways containing three lanes in each highway with both upward and downward traffic patterns. Table 1 represents the values of various parameters used in this experiment for testing purposes only. The FANET simulation presented in this work consists of 50 UAVs with CBR and TCP sources on four traffic patterns under varying loads. The size of the packet is kept fixed to 512 bytes because it is found to be adequate for bulk packets that are generally supported by high-speed devices [63]. The length of the highway is set to 1 km., with a maximum speed of UAVs is fixed in the range of 0-60 m/s. There may be chances that the source-destination pair may be in the same or different lane on highways. As stated in section 2, this simulation aims to analyze the performances of AODV and DSR reactive routing protocols for the FANET environment. A simulation time of 900 seconds is set to observe the behavior of UAVs. However, it is not optimum; generally, the simulation time of at least 300 seconds is sufficient to observe the dynamics of vehicular networks [64]. It is observed that the density of UAVs varies both in time and space during these 900 seconds.

5.1 Performance Metrics

Various metrics to measure the network performance as described in RFC2501 [65] are utilized to analyze the performance of routing protocols in FANET environment. In this paper, for analyzing the performance of both reactive routing protocols (AODV and DSR), especially in highway mobility model, six famous performance metrics viz. packet delivery fraction, normalized routing load, routing overhead, average end-to-end delay, packet loss, and throughput were taken into consideration that are briefed in upcoming subsections.

Table 1: Simulation Parameters

<i>Parameters</i>	<i>Value</i>
Routing Protocols	AODV, DSR
No. of UAVs	50
MAC Layer	802.11P
Data Traffic	CBR, TCP
No. of Sources (Load)	10 (Low), 40 (High)
Packet Size	512 bytes
Highway Length	1 km.
Mobility Model	Highway Mobility Model
No. of Highways	2
No. of Lanes (Each Highway)	3
Maximum Speed	0-60 m/sec (interval of 10)
Pause Time	0 sec.
Simulation Time	900 sec.

5.1.1 Packet Delivery Fraction (PDF)

PDF provides the ratio of total packets transmitted by a UAV source to the total number of packets received a UAV sink as given in (7). It is calculated by dividing the total amount of packets received by a destination node by the total amount of packets sent by the source node. Higher values of PDF indicate better performance of a network which is generally desired in all situations.

$$PDF = \frac{Num_Packets_Recv}{Num_Packets_Trans} \times 100 \quad (7)$$

5.1.2 Normalized Routing Load (NRL)

NRL indicates the fraction of total control packets transmitted with respect to a data packet delivered at the destination node as given in (8). Packet transmissions are usually counted in a hop-wise manner. It should not only include all bits of control packets but also header bits of data packets as well for calculating the bit efficiency [29].

$$NRL = \frac{Control_Packets_Sent}{Data_Packet_Recv} \quad (8)$$

5.1.3 Routing Overhead (ROH)

ROH indicates the fraction of total control packets sent by total data packets received successfully as given in (9). The lower amount of ROH shows network stability when new routes are being discovered.

$$ROH = \frac{Total_Control_Packets_Sent}{Total_Data_Packets_Recv} \quad (9)$$

5.1.4 Average End-to-End Delay

It is defined as the average time taken by packets from the source node to the destination node. Generally, the time taken in discovering routes and waiting in a queue is also considered for this metric. For real-time traffic monitoring using FANET environment, very less or no delay is desired. Delay can be calculated as given in (10), whereas average delay can be computed using (11) as follows:

$$Delay = (Recv_{Time} - Sending_{Time}) \quad (10)$$

$$Avg_{Delay} = \frac{Total_Delay}{Total_Packets} \quad (11)$$

5.1.5 Packet Loss

As the name indicated, it is a count of the total number of packets lost during transmission. Higher packet loss represents the low performance of a network. Generally, there may be several reasons for packet loss, i.e., network congestion, issues with source or destination hardware/ infrastructure, security threats, etc.

$$Packet_{Loss} = Total_Packets_Dropped \quad (12)$$

5.1.6 Throughput (TP)

Throughput is the measurement of the amount of data transmitted in an interval of time. Generally, it is used to measure the capacity of a network. The time interval is usually measured in milliseconds but

can be calculated for any time metric as desired. The general formula for computing the throughput of a network is given in (13) as follows:

$$TP = \frac{Total_Bits_Trans}{Time} \quad (13)$$

Several factors can affect the throughput, i.e., unreliable communication link, frequent changes in topology, low bandwidth, etc. For calculating the overall throughput of the FANET environment, the throughput of the whole data can be aggregated.

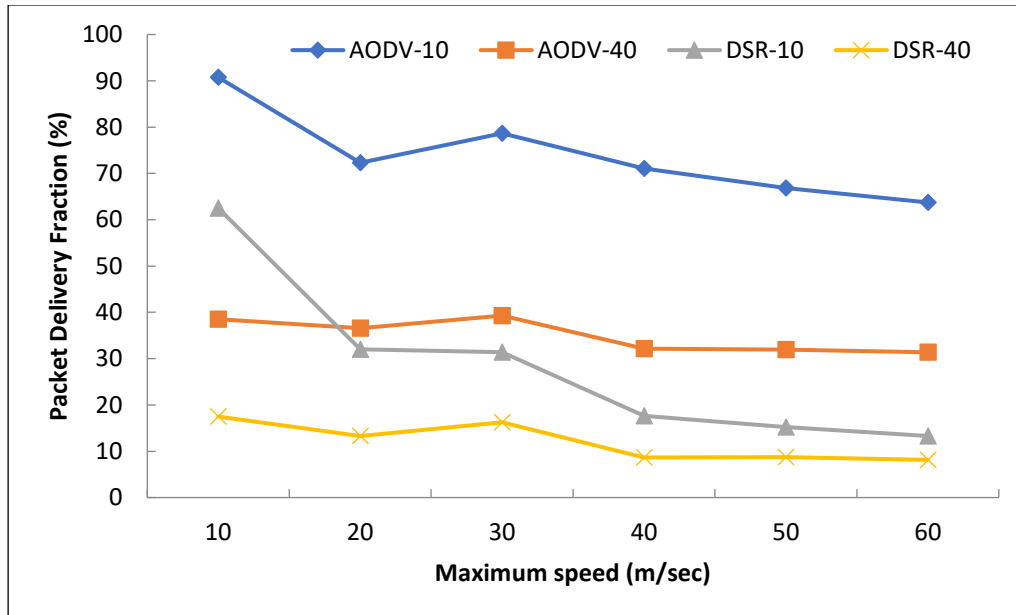
6. Results and Discussion

The performance of both routing protocols has been evaluated using experimental analysis based on the simulation setup and performance metrics as discussed in section 5. Metric wise performance of both AODV and DSR routing protocols for highway mobility is discussed as follows:

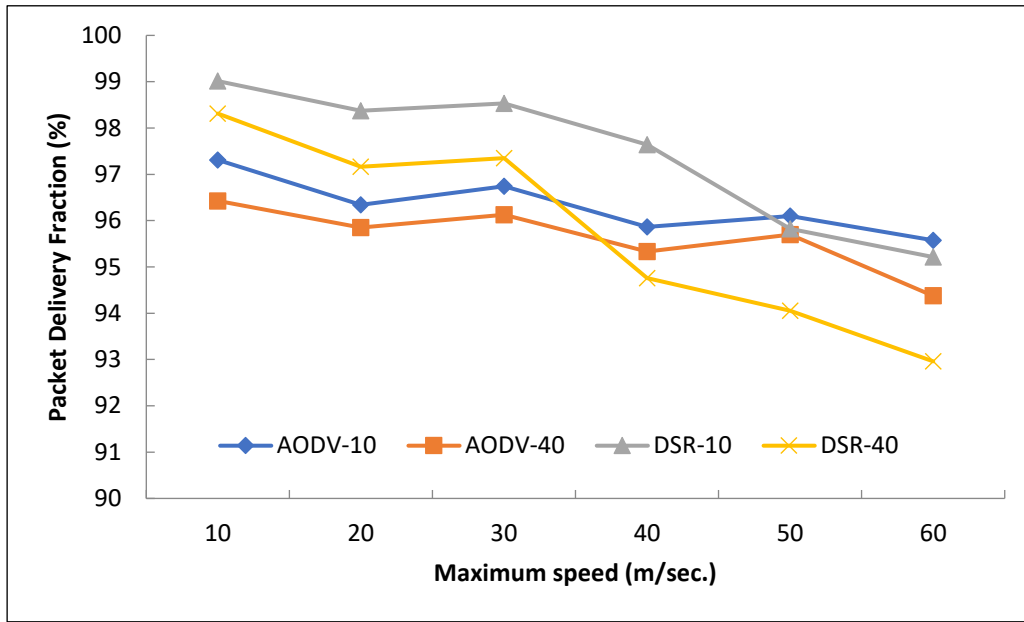
6.1 Packet Delivery Fraction

As mentioned in section 4, CBR and TCP traffic sources have been used to generate random traffic for evaluating the behavior of routing protocols in the FANET environment. Performance comparison of both protocols concerning CBR and TCP traffic sources is represented in figure 6 (a) and (b) respectively.

For CBR, AODV protocol was able to achieve a PDF of 90 when the speed of nodes was slow (i.e., 10m/s) and the network load is also low (10), but when the speed of nodes increases (i.e., 60 m/s) PDF starts decreasing and reaches to 64 only. Comparatively, DSR was able to achieve a PDF of only 13 to 62 in case of high and slow speed, respectively in the same environment when the network load was ten only. On the other hand, when the number of sources increased to 40, AODV was able to achieve the PDF of 39 during slow speed and 30 in case of high speed, whereas DSR performed the PDF of only 17 and 10 in the same scenarios.



(a)



(b)

Fig. 6(a): PDF achieved by AODV and DSR routing protocols on CBR traffic sources

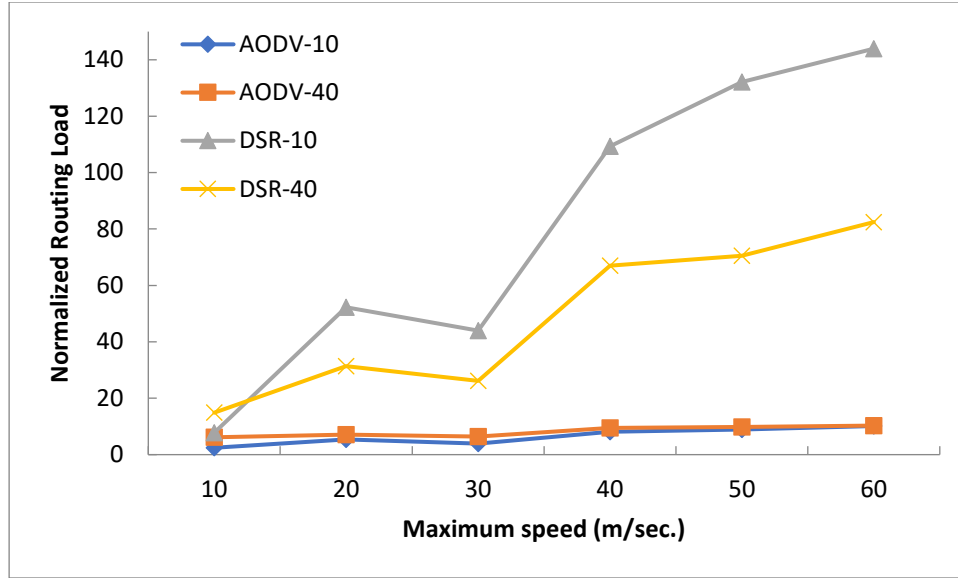
(b): PDF achieved by AODV and DSR routing protocols on TCP traffic sources

For TCP, AODV protocol was able to achieve a PDF of 97.2 when the speed of nodes was slow (i.e., 10m/s) and the network load is low (10), but when the speed of nodes increases (i.e., 60 m/s) PDF starts decreasing and reaches to 95.5 only. Comparatively, DSR was able to achieve a higher PDF of 99 in slow speed and 95 in case of high and slow speed in the same environment when the number of sources was ten only. On the other hand, when network load increased to 40, AODV was able to achieve the PDF of 96.5 during slow speed and 94.4 in case of high speed, whereas DSR performed a higher PDF of 98.5 as compared to AODV in case of slow speed and low PDF of 93 then AODV during the high speed of nodes.

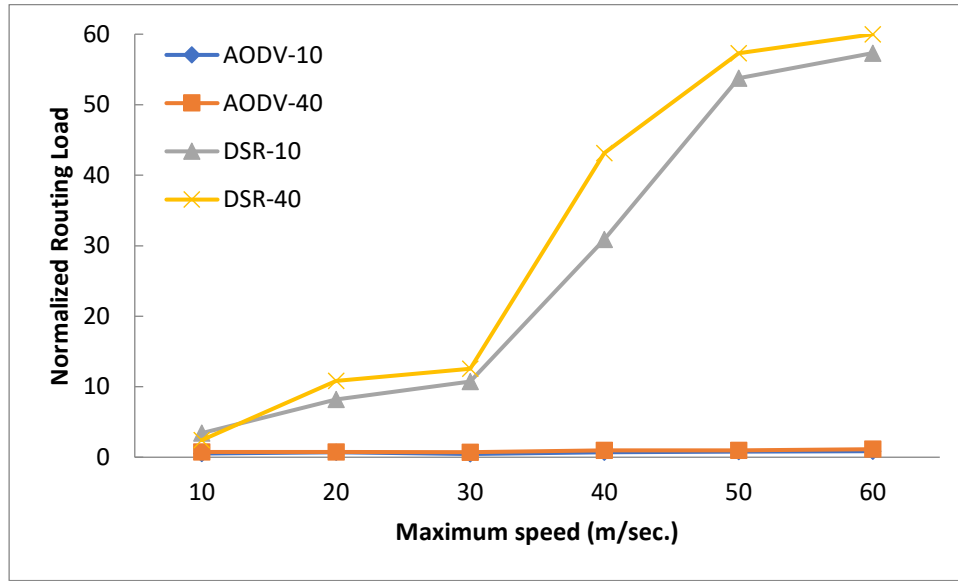
6.2 Normalized Routing Load

In the case of both TCP and CBR traffic sources, the minimum value of normalized routing load under the FANET environment was reported by AODV protocol as compared to DSR. AODV outperforms DSR not only under the varying speed of nodes but also under high and low network load. The performance comparison of both protocols for CBR and TCP traffic sources is shown in figures 7 (a) and (b) respectively.

The normalized load reported by AODV in the case of TCP traffic has been observed to zero as compared to 60 by DSR. From these experimental results, it can be easily stated that amongst both protocols, AODV is more suitable for the FANET environment.



(a)

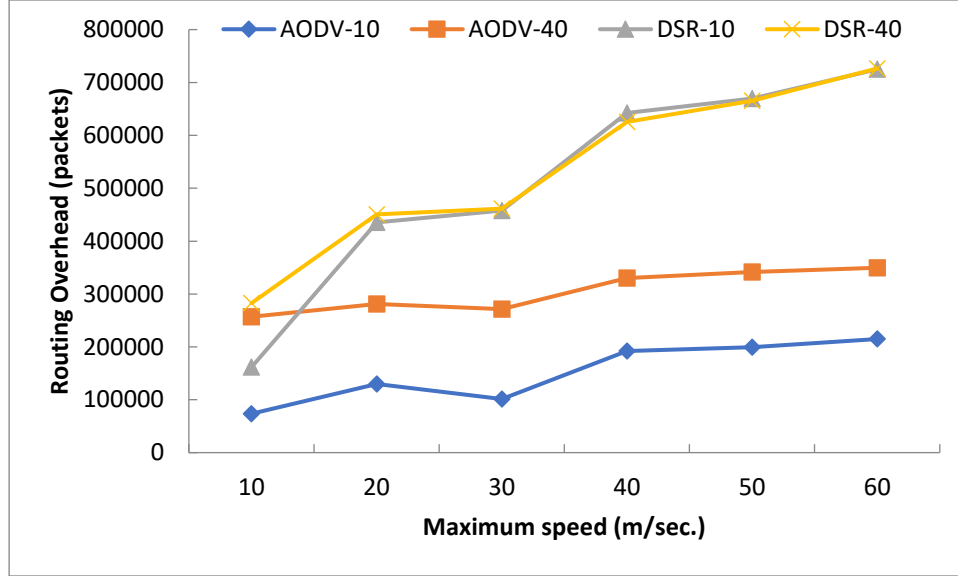


(b)

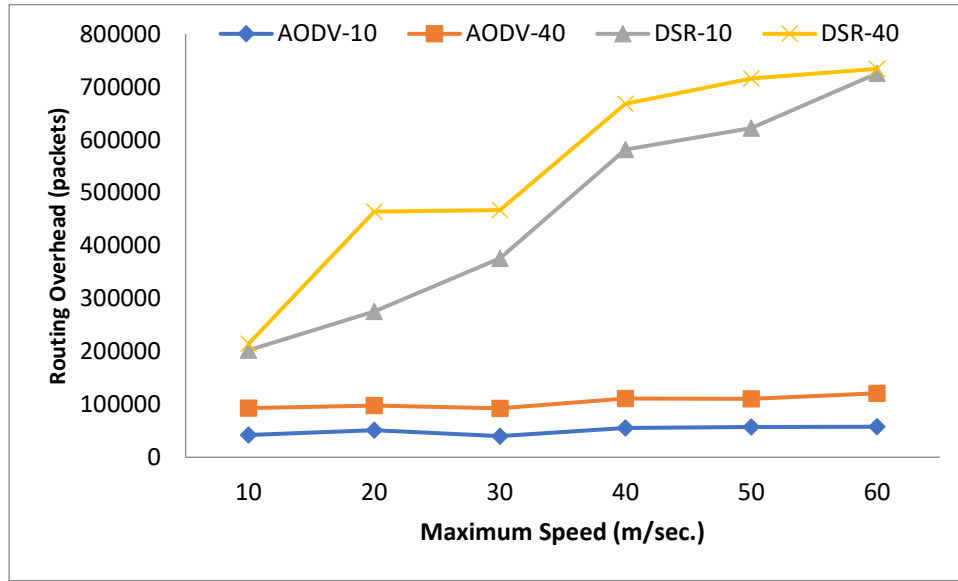
Fig.7 (a): NRL achieved by AODV and DSR routing protocols on CBR traffic sources
(b): NRL achieved by AODV and DSR routing protocols on TCP traffic sources

6.3 Routing Overhead

AODV reported the lowest routing overhead in the case of both CBR and TCP traffic sources, as shown in figure 7 (a) and (b) respectively. Generally, the DSR route replies contribute to some extent for higher routing overhead. It has been observed that with the increment in both speeds of UAVs as well as network load, the routing overhead in case of CBR increases 14 times as compared to 6 times in the case of TCP for both protocols under the same circumstances.



(a)



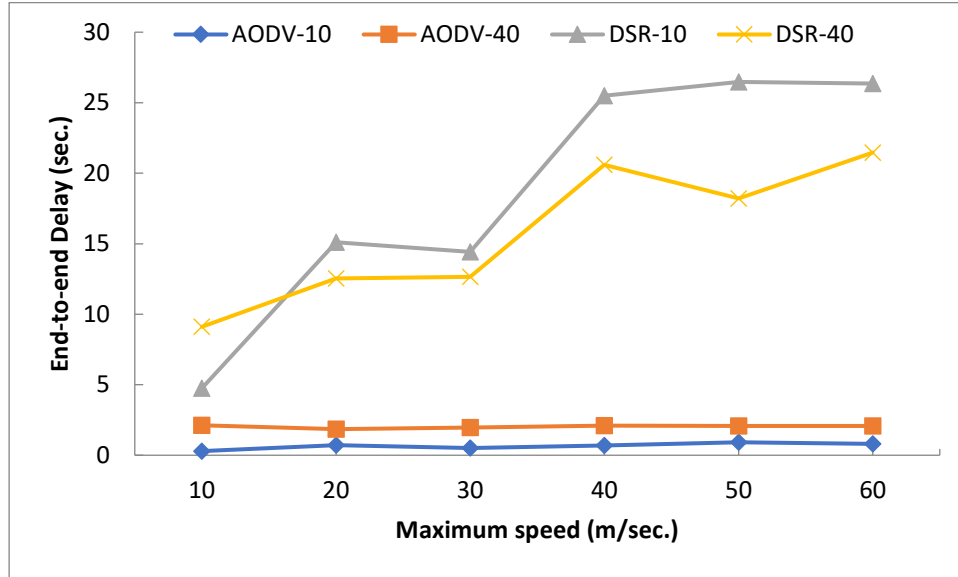
(b)

Fig. 7 (a): Routing overhead generated by AODV and DSR routing protocols on CBR traffic sources
 (b): Routing overhead generated by AODV and DSR routing protocols on TCP traffic sources

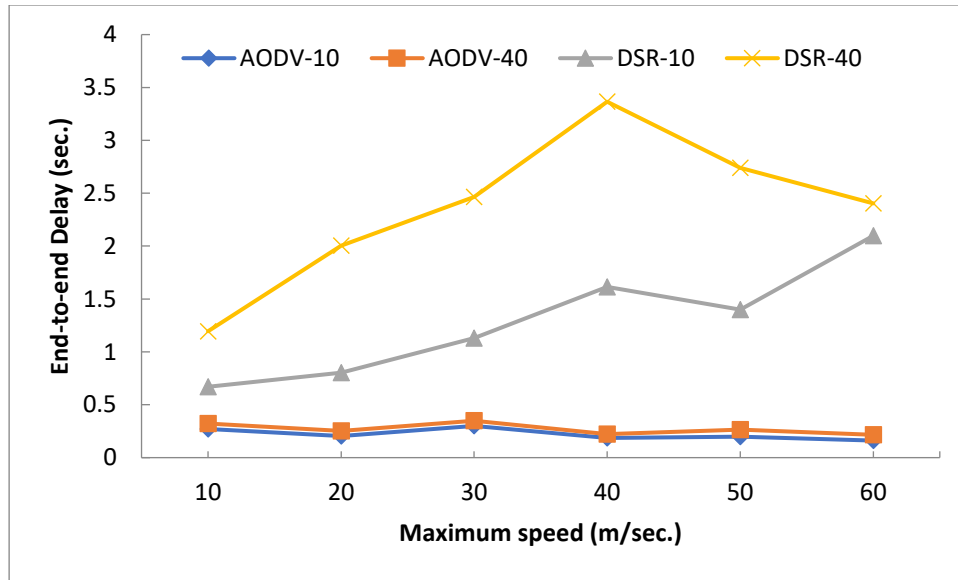
6.4 Average end-to-end delay

As mentioned in the previous section, very low delay is preferred for high performance. In this regard, observations based on experimental evaluations depicted significantly less delay by AODV on both CBR and TCP traffic sources, as shown in figure 8 (a) and (b) respectively. AODV performs better in almost all circumstances, i.e., on a varying number of traffic sources (loads) as well as the speed of UAVs. DSR reports more end-to-end delays due to route caching. Based on this analysis, it can be

stated that AODV should be preferred for FANET environments where real-time packet transmission is in demand.



(a)



(b)

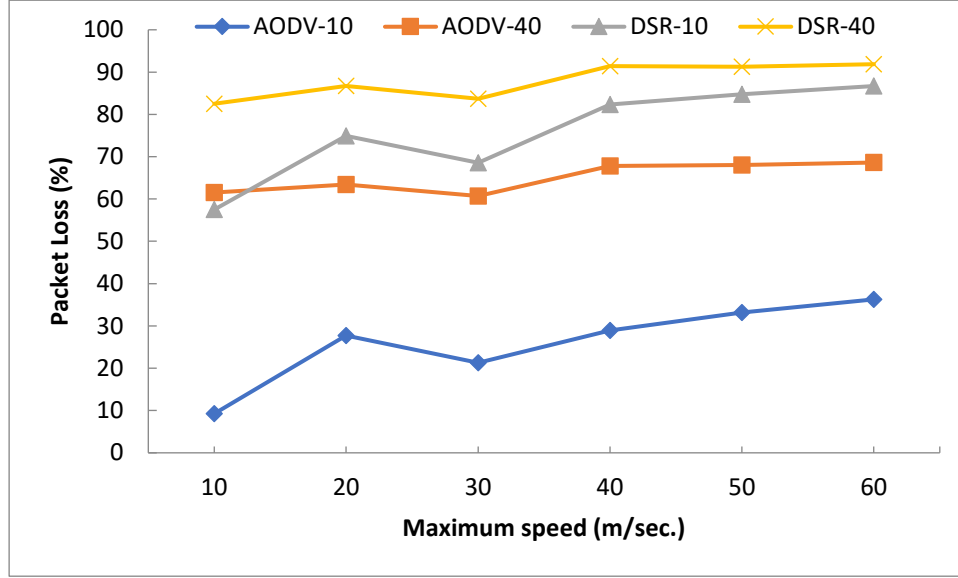
Fig. 8 (a): Average end-to-end delay by AODV and DSR routing protocols on CBR traffic sources

(b): Average end-to-end delay by AODV and DSR routing protocols on TCP traffic sources

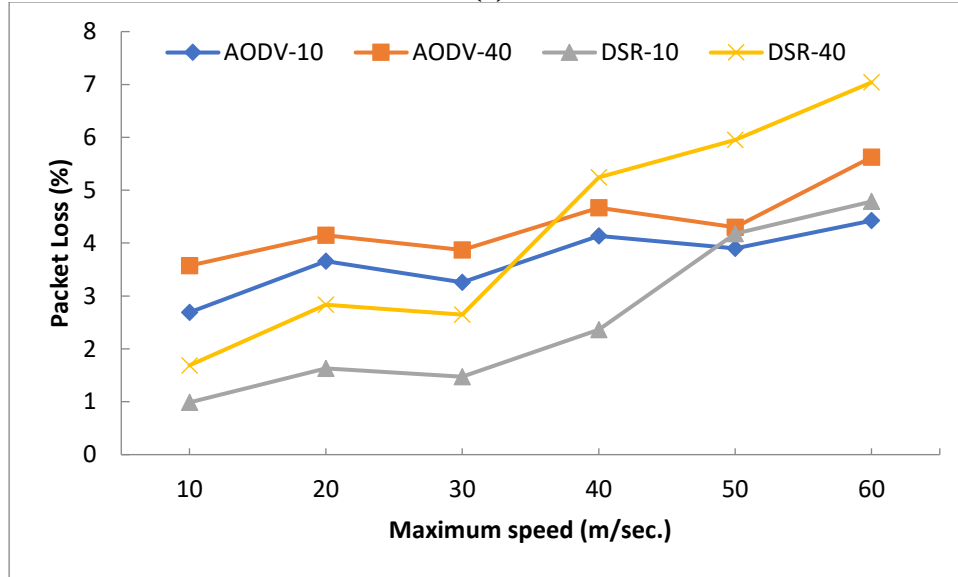
6.5 Packet Loss

As discussed in the previous section, there can be several reasons for packet losses. When evaluating the performance of both protocols in the FANET environment regarding packet loss, it has been observed that a more significant number of packets were lost in the case of DSR protocol as compared to AODV on both traffic sources as shown in figure 9 (a) and (b) respectively. *Generally, this happens because, the routing error is propagated back to the source node in the case of a link, and the process*

continues for every transmitted packet. Based on experimental evaluations of both protocols under varying network loads as well as speed, it can be stated that AODV is preferred over DSR for FANET environments.



(a)



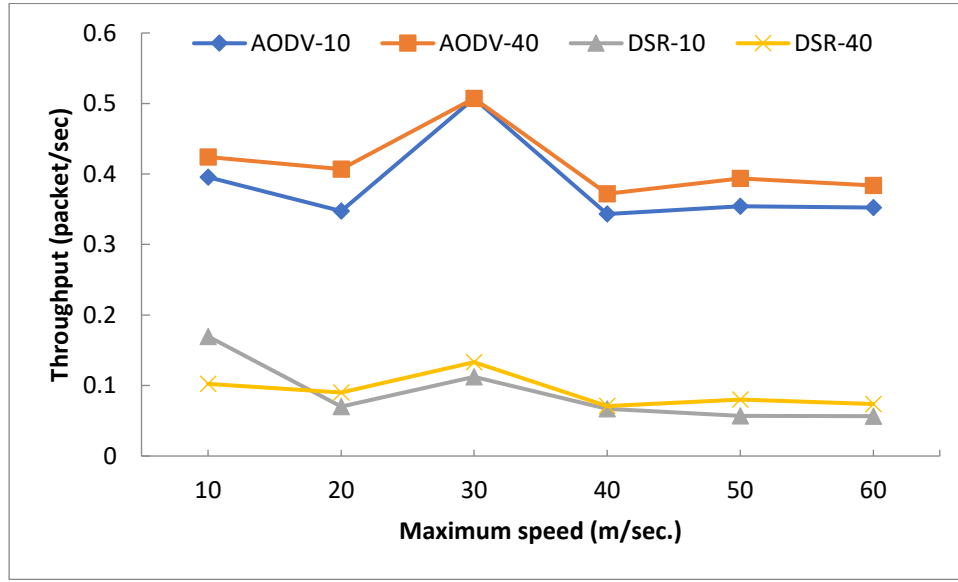
(b)

Fig. 9 (a): Packet loss by AODV and DSR routing protocols on CBR traffic sources
(b): Packet loss by AODV and DSR routing protocols on TCP traffic sources

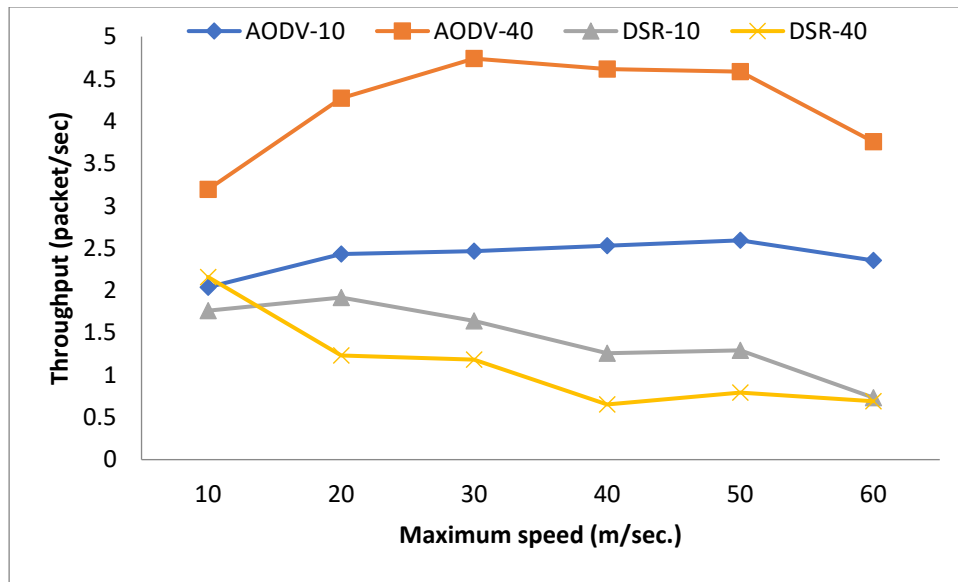
6.6 Throughput

As expected, AODV reported higher throughput as compared to DSR on both traffic sources viz. CBR and TCP. Under varying load and node speed, AODV performs better than DSR on both traffic sources, as shown in figures 10 (a) and (b) respectively. However, on TCP source, when network load, as well as speed increases, a slight decrement in throughput, has been observed in the case of AODV

protocol. Based on the experimental evaluation, high performance on behalf of throughput has been observed by AODV protocol, which makes it preferable over DSR for the FANET environment.



(a)



(b)

Fig.10 (a): Throughput obtained by AODV and DSR routing protocols on CBR traffic sources
(b): Throughput obtained by AODV and DSR routing protocols on TCP traffic sources

6.7 Summary of experimental results

As stated earlier, the goal of these experimental evaluations is to find the suitable routing protocol for Flying Ad-hoc Networks under the highway mobility model. Table 2 represents the overall performance of both protocols under varying network loads on CBR and TCP traffic sources.

Table 2: Performance comparison of AODV and DSR routing protocols under varying network loads on CBR and TCP

Network Load	Performance Metric	CBR		TCP	
		AODV	DSR	AODV	DSR
Low (10 sources)	Packet delivery fraction	✓	×	×	✓
	Normalized routing load	✓	×	✓	×
	Routing overhead	✓	×	✓	×
	Average end-to-end delay	✓	×	✓	×
	Packet Loss	✓	×	✓	×
	Throughput	✓	×	✓	×
High (40 sources)	Packet delivery fraction	✓	×	×	✓
	Normalized routing load	✓	×	✓	×
	Routing overhead	✓	×	✓	×
	Average end-to-end delay	✓	×	✓	×
	Packet Loss	✓	×	✓	×
	Throughput	✓	×	✓	×

Table 3: Summary of experimental results indicating the performances of AODV and DSR routing protocols based on CBR traffic source under varying load and speed of UAVs

Metric	Speed	10	20	30	40	50	60
Packet Delivery Fraction	AODV-10	90.74	72.31	78.68	71.07	66.86	63.73
	AODV-40	38.49	36.56	39.29	32.17	31.95	31.38
	DSR-10	62.50	32.04	31.40	17.65	15.24	13.30
	DSR-40	17.48	13.28	16.27	8.62	8.72	8.13
Normalized Routing Load	AODV-10	2.44	5.40	3.89	8.14	8.96	10.15
	AODV-40	6.16	7.10	6.39	9.48	9.86	10.30
	DSR-10	7.79	52.24	43.97	109.33	132.15	143.95
	DSR-40	14.89	31.34	26.13	66.97	70.50	82.46
Routing Overhead	AODV-10	73524	129697	101633	192238	199126	214847
	AODV-40	257005	280970	271588	330290	341503	349610
	DSR-10	161604	435238	458055	642650	669758	724982
	DSR-40	282623	450765	461163	625606	665225	726503
End-to-End Delay	AODV-10	0.29	0.72	0.51	0.70	0.92	0.81
	AODV-40	2.12	1.85	1.96	2.10	2.08	2.08
	DSR-10	4.75	15.10	14.42	25.49	26.47	26.35
	DSR-40	9.11	12.54	12.64	20.60	18.21	21.46
Packet Loss	AODV-10	9.26	27.69	21.32	28.93	33.14	36.27
	AODV-40	61.51	63.44	60.71	67.83	68.05	68.62
	DSR-10	57.50	74.96	68.60	82.35	84.76	86.70
	DSR-40	82.52	86.72	83.73	91.38	91.28	91.87
Throughput	AODV-10	0.40	0.35	0.51	0.34	0.35	0.35
	AODV-40	0.42	0.41	0.51	0.37	0.39	0.38
	DSR-10	0.17	0.07	0.11	0.07	0.06	0.06
	DSR-40	0.10	0.09	0.13	0.07	0.08	0.07

From table 2, it can be stated that AODV should be preferred for achieving better routing in FANET environments. The detailed values of experimental evaluations under CBR and TCP traffic sources are

depicted in tables 3 and 4 respectively. For analyzing the performance of both routing protocols, the speed of UAVs has been kept in a range of 10-60 m/sec under varying loads (10 and 40).

Table 4: Summary of experimental results indicating the performance of AODV and DSR routing protocols based on TCP traffic source under varying load and speed of UAVs

Metric	Speed	10	20	30	40	50	60
Packet Delivery Fraction	AODV-10	97.31	96.34	96.74	95.87	96.10	95.58
	AODV-40	96.43	95.85	96.13	95.33	95.70	94.37
	DSR-10	99.01	98.37	98.53	97.64	95.82	95.21
	DSR-40	98.31	97.16	97.35	94.76	94.05	92.96
Normalized Routing Load	AODV-10	0.52	0.69	0.46	0.68	0.78	0.86
	AODV-40	0.75	0.75	0.72	0.98	0.97	1.14
	DSR-10	3.42	8.20	10.73	30.88	53.77	57.32
	DSR-40	2.43	10.84	12.55	43.15	57.32	60.00
Routing Overhead	AODV-10	42347	50965	39940	55303	57456	57639
	AODV-40	92986	97474	92543	110824	110623	120910
	DSR-10	202267	275363	375880	582098	622600	725793
	DSR-40	214317	464279	467234	668676	715945	734513
End-to-End Delay	AODV-10	0.27	0.20	0.30	0.19	0.20	0.16
	AODV-40	0.32	0.25	0.35	0.22	0.26	0.22
	DSR-10	0.67	0.80	1.13	1.61	1.40	2.10
	DSR-40	1.19	2.01	2.46	3.37	2.74	2.40
Packet Loss	AODV-10	2.69	3.66	3.26	4.13	3.90	4.42
	AODV-40	3.57	4.15	3.87	4.67	4.30	5.63
	DSR-10	0.99	1.63	1.47	2.36	4.18	4.79
	DSR-40	1.69	2.84	2.65	5.24	5.95	7.04
Throughput	AODV-10	2.04	2.43	2.46	2.53	2.59	2.36
	AODV-40	3.19	4.27	4.74	4.61	4.59	3.76
	DSR-10	1.76	1.92	1.64	1.26	1.29	0.73
	DSR-40	2.16	1.23	1.18	0.65	0.79	0.69

7. Conclusion

One of the most challenging tasks in flying ad-hoc networks is the communication among UAVs that are flying at varying speeds and mobility. The significant contribution of the work is to present the monitoring of traffic using the highway mobility model and performance analysis of two well-known UAV routing algorithms viz. AODV and DSR on several parameters under varying network conditions i.e., speed, load, and mobility. Traffic simulations are accomplished with the help of CBR and TCP sources using NS-2 simulator for highway mobility model. Experimental evaluations based on six well-known performance metrics viz. packet delivery fraction, end-to-end delay, packet loss, routing overhead, normalized routing load, and throughput indicated the better performance of AODV as

compared to DSR in almost all circumstances. Throughput is one of the important metrics that is observed to be higher in the case of the AODV routing protocol. Average throughput on varying speed of nodes on CBR traffic source is observed to be 0.38 of AODV-10 as compared to 0.09 of DSR-10, whereas 0.41 of AODV-40 with 0.09 of DSR-40. On the other hand, the average throughput on varying speed of nodes on TCP traffic source is observed to be 2.40 of AODV-10 as compared to 1.43 of DSR-10, whereas 4.19 of AODV-40 as compared to 1.12 of DSR-40. In all circumstances, approximately 63% more throughput is observed using the AODV routing protocol as compared to the DSR protocol. In the future, performance analysis of other routing protocols for the FANET environment which is based on different mobility models and varying traffic conditions could also be carried out.

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